ML Algo $\left\{ \begin{array}{l} \text{classifier} \\ \text{regressor} \end{array} \right\}$ feature dist metric

f

$f_1, f_2, \ldots f_n$

$\text{dist} = |f_i - f_j|$

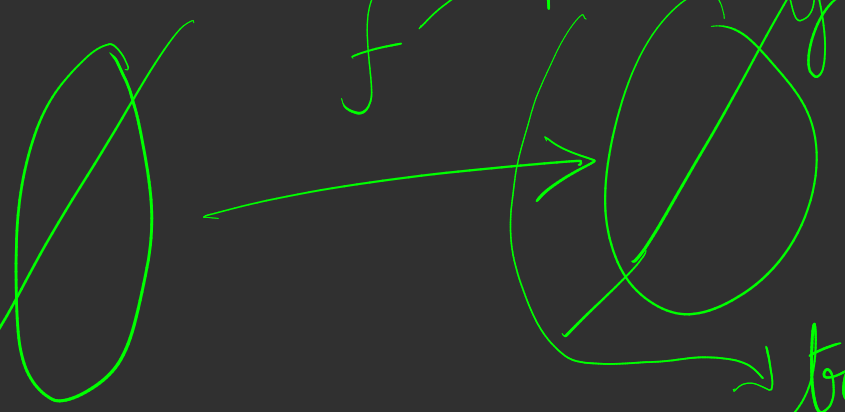
data

X

trn

tst

f $[X, Y]$



$f \rightarrow$ parameters

Y target

f_{params}

trn

tst

$\hat{f}$

test

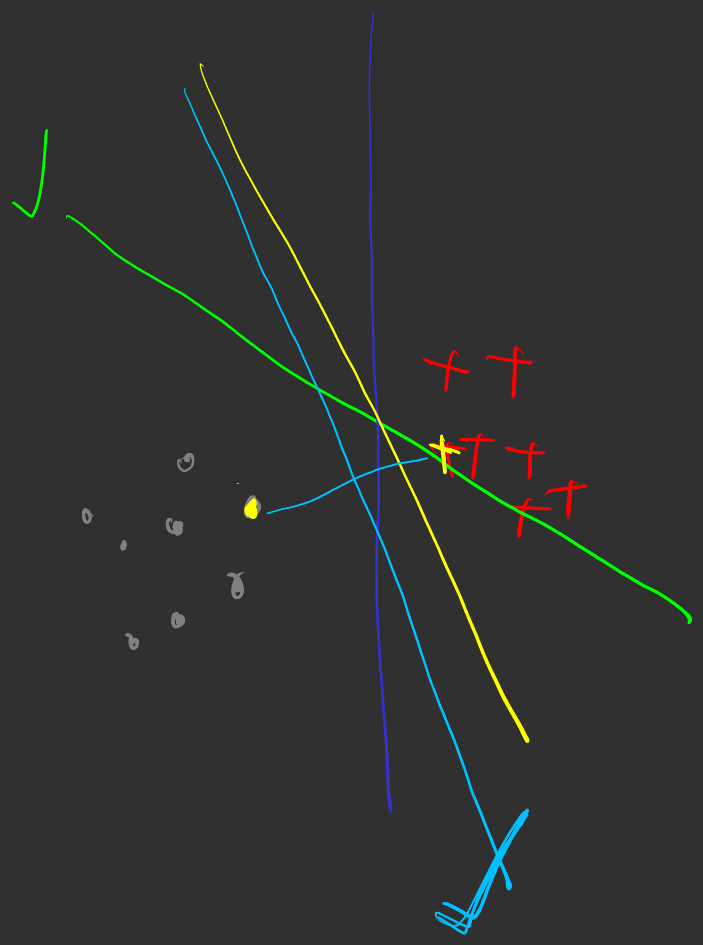
Scaling $\left\{ \begin{array}{l} \mu, \sigma \\ m, M \end{array} \right.$

$$err \; \frac{1}{N}\left|y - \hat{y}\right| \quad \text{MAE}$$

$\hookrightarrow \geq 0$

$$\sqrt{\frac{1}{N}\left(y - \hat{y}\right)^2} \quad \text{RMSE}$$

0-1



$W^T \bar{X} + b$

non-linear

Kernel trick

$W^T X + b$

$f_1 \; f_2 \; f_3$

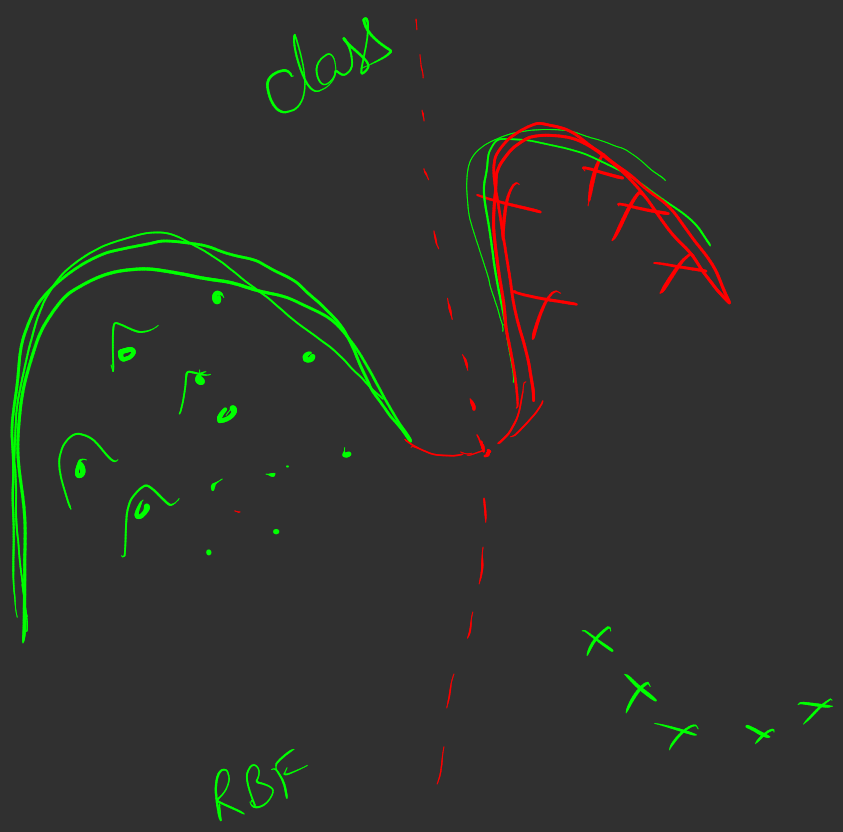
$f_1 \; f_1^2, \; f_1 f_2$

regularization noise X

$w^T x + b$

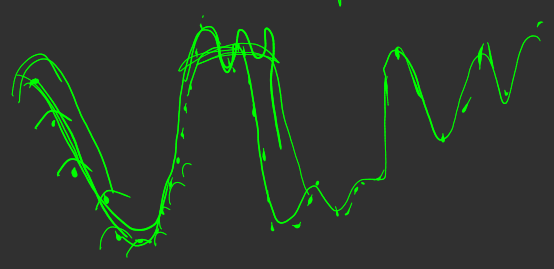
X

class



RBF

poly



$a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$